

# Advanced Mathematical Physics – Project Specifications

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## Objective

Extend a topic from the course and apply it to a meaningful problem in physics.

## Topic

- Must extend the course material
- Must include a clear physics application
- Should demonstrate mathematical depth and physical understanding

## Examples (for guidance only)

- Banach space
- $C^*$ -algebra
- Conformal mapping
- Sensitive convergence series tests

## Report ( $\approx 10$ pages)

Include:

- Introduction (problem and motivation)
- Mathematical framework
- Analysis / solution
- Physical interpretation
- Conclusion
- References

Ensure clear writing, proper formatting, and well-explained equations/figures.

## Presentation

- 10–15 minutes
- Explain the problem, key methods, results, and physical meaning
- Focus on clarity

## Evaluation

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| <b>Project (Written Report) 10%:</b> <ul style="list-style-type: none"><li>- Relevance to course material</li><li>- Mathematical rigor</li><li>- Quality of physical application</li><li>- Clarity and organization</li></ul> | <b>Oral Presentation 10%:</b> <ul style="list-style-type: none"><li>- Clarity of explanation</li><li>- Understanding of the topic</li><li>- Organization and delivery</li></ul> |
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